MASTER LIST

Repeated Topics: The Complete Union

One list combining the question bank matches, the questions repeated across the Dec 2025 and Jun 2026 papers, and the questions common to all three.

S.E. Sem III | Embedded Systems | D Y Patil University

9	25	18	5
TOPICS	QB QUESTIONS	EXAM QUESTIONS	IN ALL THREE
Repeated in both papers	17 direct, 8 related	Dec 2025 and Jun 2026	QB, Dec and Jun

These nine topics account for **51 of the 80 marks** offered in Dec 2025 and **52 of the 80 marks** offered in Jun 2026.

HOW TO READ THE TAGS

ALL THREE	Question bank question that was also asked in both papers. Study first.
TOPIC REPEAT	Same topic in both papers, but a different question each time. The QB has several matching questions.
NOT IN QB	Asked in both papers, but the question bank has no direct question. Prepare from notes.
DIRECT / RELATED	Direct is a QB question on the same topic. Related is a close question on a neighbouring topic.

PRIORITY ORDER

What to Study First

Every repeated topic, ranked by how strongly it recurs

TOPIC	QUESTION BANK	DEC 2025	JUN 2026
Tier 1: in the question bank and both papers			
Multiplexer and De-Multiplexer	M1 Q9, M1 Q10	Q2a (5)	Q3a (10)
Number system conversion	M1 Q1, M1 Q2, M1 Q3	Q1a (4)	Q2a (5)
Microprocessor vs Microcontroller	M2 Q9	Q1b (4)	Q2b (5)
Washing Machine case study	M5 Q3	Q1d (4)	Q3b (10)
DC motor interfacing with 8051	M5 Q4, M5 Q9	Q2e (5)	Q1d (4)
Tier 2: same topic every time, new question each time			
Flip-flops	M1 Q11, M1 Q12, M1 Q13	Q3a (10)	Q1a (4)
ARM: pipelining and CPSR	M6 Q5, M6 Q8, M6 Q3	Q1e (4)	Q1e (4)
ARM: blocks and operating modes	M6 Q2, M6 Q4	Q2f (5)	Q2f (5)
Tier 3: in both papers, missing from the question bank			
Internal memory organization of 8051	none (closest: M3 Q1, M3 Q2, M2 Q11)	Q3b (10)	Q2c (5)

Question bank checklist (direct matches, 17 questions): M1 Q9, M1 Q10, M1 Q1, M1 Q2, M1 Q3, M1 Q11, M1 Q12, M1 Q13, M2 Q9, M5 Q3, M5 Q4, M5 Q9, M6 Q5, M6 Q8, M6 Q3, M6 Q2, M6 Q4.

BY MODULE

Full Union, Module-wise

Question bank questions, then the Dec 2025 and Jun 2026 exam questions for each topic

MODULE 01

Number Systems & Digital Logic

Multiplexer and De-Multiplexer

Dec 2025 Q2a (5 marks) | Jun 2026 Q3a (10 marks)

ALL THREE

QB M1 Q9	What is Multiplexer? Draw a 4:1 Multiplexer, write down the Boolean equation and design it with logic gates.	DIRECT
QB M1 Q10	Explain the working of 4:1 Demultiplexer. Design it with logic gates.	ALL THREE
DEC Q2a	Describe the working of De-Multiplexer with diagram and circuit.	EXAM
JUN Q3a	Illustrate the Multiplexer and De-Multiplexer with circuit diagram.	EXAM

Prepare both together. Jun asked Mux and De-Mux in one 10-mark question.

Number system conversion

Dec 2025 Q1a (4 marks) | Jun 2026 Q2a (5 marks)

ALL THREE

QB M1 Q1	Convert $(73.301)_{10}$ into binary, octal and hexadecimal equivalent.	DIRECT
QB M1 Q2	Convert: $(BC)_{16}$ to decimal; $(2314)_8$ to decimal; $(1000011)_2$ to decimal; $(647)_{10}$ to hexadecimal.	ALL THREE
QB M1 Q3	Express in decimal: $(10110.0101)_2$, $(16.5)_{16}$, $(26.24)_8$.	DIRECT
DEC Q1a	Perform the conversion of an octal number $(251)_8$ into decimal number.	EXAM
JUN Q2a	Evaluate the conversion of decimal number (625) to its Hexadecimal equivalent number.	EXAM

M1 Q2 covers both types asked so far: octal to decimal and decimal to hex.

Flip-flops

Dec 2025 Q3a (10 marks) | Jun 2026 Q1a (4 marks)

TOPIC REPEAT

QB M1 Q11	Draw the circuit of SR Latch and explain its working with truth table.	DIRECT
QB M1 Q12	Draw the circuit of D flip-flop and explain its working with truth table.	DIRECT
QB M1 Q13	Draw the circuit of SR flip-flop and explain its working with truth table.	DIRECT
DEC Q3a	Illustrate the working of RS Flip-Flop with neat diagram and truth table.	EXAM
JUN Q1a	Explain the working of D-type Flip-Flop with circuit and truth table.	EXAM

Dec asked the RS flip-flop and Jun asked the D flip-flop. The SR latch has not been asked yet.

MODULE 02

Embedded System Fundamentals

Microprocessor vs Microcontroller

Dec 2025 Q1b (4 marks) | Jun 2026 Q2b (5 marks)

ALL THREE

QB M2 Q9	Compare Microprocessors and Microcontrollers.	ALL THREE
QB M2 Q1	What is an embedded system? How does it differ from a general-purpose computer?	RELATED
DEC Q1b	Compare Microprocessor and Microcontroller.	EXAM
JUN Q2b	Compare Microprocessor and Microcontroller.	EXAM

Word-for-word repeat in both papers. M2 Q1 is a similar comparison question.

MODULE 03

8051 Architecture & Instruction Set

Internal memory organization of 8051

Dec 2025 Q3b (10 marks) | Jun 2026 Q2c (5 marks)

BOTH PAPERS, NOT IN QB

QB M3 Q1	Draw the architecture of 8051 and explain in detail.	RELATED
QB M3 Q2	Draw the block diagram of 8051 and explain the main blocks.	RELATED
QB M2 Q11	What are the different memory types used in embedded systems?	RELATED
DEC Q3b	Explain the Internal RAM organization of 8051 and explain it.	EXAM
JUN Q2c	Explain the internal memory organization of 8051 microcontroller.	EXAM

No QB question asks this directly. Study internal RAM (register banks, bit-addressable area, scratch pad) and SFRs from your 8051 notes.

MODULE 05

Interfacing & Embedded Case Studies

Washing Machine case study

Dec 2025 Q1d (4 marks) | Jun 2026 Q3b (10 marks)

ALL THREE

QB M5 Q3	Discuss in detail the Embedded case study of "Washing Machine".	ALL THREE
QB M5 Q2	Discuss in detail the Embedded case study of "Digital Camera".	RELATED
DEC Q1d	Investigate the case study of Washing Machine in Embedded Systems.	EXAM
JUN Q3b	Formulate and describe a case study of washing machine.	EXAM

Marks rose from 4 to 10, so prepare a full-length answer. Digital Camera follows the same format.

DC motor interfacing with 8051

Dec 2025 Q2e (5 marks) | Jun 2026 Q1d (4 marks)

ALL THREE

QB M5 Q4	Draw detailed diagram of DC Motor interfacing with 8051 microcontroller and write suitable program for the same.	ALL THREE
QB M5 Q9	Switch SW is connected to P2.0; P1.0 and P1.1 are the two inputs of a DC Motor. If SW = 1 rotate clockwise, if SW = 0 reverse the direction. Write the program.	DIRECT
QB M5 Q5	Draw detailed diagram of Stepper Motor interfacing with 8051 microcontroller and write suitable program for the same.	RELATED
QB M5 Q8	Explain what is "Wave Drive, Full Step and Half Step" sequence with reference to stepper motor interfacing.	RELATED
DEC Q2e	Draw the diagram for DC motor interfacing with 8051 and explain it with simple program.	EXAM
JUN Q1d	Discuss DC motor interfacing with 8051 Microcontroller with diagram.	EXAM

M5 Q9 is the ready-made program for the DC motor diagram.

MODULE 06

ARM Architecture & Cortex Processors

ARM: pipelining and CPSR

Dec 2025 Q1e (4 marks) | Jun 2026 Q1e (4 marks)

TOPIC REPEAT

QB M6 Q5	Explain how the pipelining process in ARM architecture works.	DIRECT
QB M6 Q8	Explain three stage pipelining process in ARM architecture.	DIRECT
QB M6 Q3	Discuss the General-Purpose Register set and CPSR in ARM architecture.	DIRECT
DEC Q1e	Describe three stage pipelining in ARM microcontroller.	EXAM
JUN Q1e	Elaborate CPSR register in ARM Microcontroller.	EXAM

Both papers asked a 4-mark ARM question in the compulsory Q1, on a different subtopic each time.

ARM: blocks and operating modes

Dec 2025 Q2f (5 marks) | Jun 2026 Q2f (5 marks)

TOPIC REPEAT

QB M6 Q2	Draw the ARM architecture and discuss the main blocks.	DIRECT
QB M6 Q4	Describe the different operating modes in ARM architecture.	DIRECT
QB M6 Q1	Explain the main features of ARM architecture.	RELATED
DEC Q2f	Draw and explain the main blocks of ARM architecture.	EXAM
JUN Q2f	Describe various operating modes in ARM Microcontroller.	EXAM

Both papers asked a 5-mark ARM question in the same slot of Q2, on a different subtopic each time.